

David Cornelis Groothuizen Dijkema

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Education

University of Auckland | Waipapa Taumata Rau

Doctoral Candidate in Mathematics Mar. 2022 — present

Thesis topic: Dynamical dependence on topology of symmetric heteroclinic networks

Supervisors: Profs. Claire Postlethwaite and Vivien Kirk

Expected submission: August 2025

Bachelor of Science (Honours) 2021

with First Class Honours in Applied Mathematics

Honours dissertation: A heteroclinic network in a spatially-extended four species interaction model

Supervisor: Prof. Claire Postlethwaite

Bachelor of Engineering (Honours) 2017 — 2020

with First Class Honours in Engineering Science

Honours dissertation: Synthetic-aperture radar processing on the NVidia Jetson AGX Xavier

Supervisors: Prof. John Cater and Assoc Prof. Nicholas Rattenbury

Publications and preprints

David C Groothuizen Dijkema and Claire M Postlethwaite. (2023). Travelling waves and heteroclinic networks in models of spatially-extended cycle competition. *Nonlinearity*, **36**(12):6546–6588.

DOI: [10.1088/1361-6544/ad0212](https://doi.org/10.1088/1361-6544/ad0212). arXiv: [2208.05630](https://arxiv.org/abs/2208.05630) [math.DS].

David C Groothuizen Dijkema, Vivien Kirk, and Claire M Postlethwaite. Analysis of dynamics near heteroclinic networks in $\mathbb{R}^4$ with a projected map. *Submitted to SIAM Journal on Applied Dynamical Systems*.

arXiv: [2410.21486](https://arxiv.org/abs/2410.21486) [math.DS].

David C Groothuizen Dijkema, Vivien Kirk, Claire M Postlethwaite, and Alastair M Rucklidge. Continuity of projected maps near heteroclinic networks in $\mathbb{R}^4$. *In preparation. Preprint available upon request.*

David C Groothuizen Dijkema, Vivien Kirk, Claire M Postlethwaite, and David J W Simpson. Strings of sausages and regular cycling near the Rock-Paper-Scissors-Lizard-Spock network. *In preparation.*

Awards and scholarships

Honourable mention for the Australia and New Zealand Industrial and Applied Mathematics T M Cherry Prize for Best Student Talk at the ANZIAM Conference 2025

New Zealand Mathematical Society Aitken Prize for Best NZMS Student Talk at the Joint Meeting of the NZMS, AustMS, and AMS 2024

New Zealand Mathematics Research Institute Kalman Summer Scholarship (To support attendance at the NZMRI 2024 Summer School)	
Nonlinearity Outstanding Reviewer Award	2023
University of Auckland Academic Career Advancement Award (To participate in the Academic Career Advancement Programme)	
Australia and New Zealand Industrial and Applied Mathematics Prize for Best Poster Presentation at the NZMS Colloquium	
Honourable mention for the New Zealand Mathematical Society Aitken Prize for Best Student Talk at the NZMS Colloquium	
Royal Society of New Zealand Te Apārangi Marsden Fund Doctoral Scholarship (Full scholarship, including stipend and fees)	2022
University of Auckland Department of Mathematics Student Research Conference prize First in Course Award for MATHS 761 (Dynamical Systems)	2021
University of Auckland Postgraduate Honours Scholarship (Full scholarship, including stipend and fees)	
University of Auckland Summer Research Scholarship	2020
First in Course Award for ENGSCI 355 (Applied Modelling in Simulation and Optimisation)	2019
University of Auckland Faculty of Engineering Dean's Honours List	2017, 2018, 2019

Teaching experience

Department of Mathematics, Faculty of Science, University of Auckland	2022 — 2024
Teaching support Developed resources to support the use of the software CoCo in MATHS 761 (Dynamical Systems), a postgraduate course. Developed resources include MATLAB templates and completed examples for student use, and tutorial worksheets.	
Graduate Teaching Assistant for COMPSCI 120 (Mathematics for Computer Science) COMPSCI 225 (Discrete Structures in Mathematics and Computer Science) MATHS 108 (General Mathematics 1) MATHS 120 (Algebra 1) MATHS 130 (Calculus 1) MATHS 253 (Algebra and Calculus 3) MATHS 260 (Differential Equations) Mathematics Assistance Room (an undergraduate drop-in clinic)	
Department of Engineering Science and Biomedical Engineering,	2020
Faculty of Engineering, University of Auckland	2020
Teaching Assistant for ENGSCI 255 (Modelling and Analytics in Operations Research) ENGSCI 211 (Mathematical Modelling 2)	

Service

Co-organiser of a minisymposium at SIAM DS 25	2025
I co-organised a two-part minisymposium titled <i>Dynamics near Homoclinic and Structures</i> with Alexander Lohse, Universität Hamburg, Germany at the SIAM Conference on Applications of Dynamical Systems 2025, Denver, CO, USA	

Chair of the Social Committee of the Joint Meeting of the NZMS, AustMS, and AMS	2024
I served as the chair of a committee composed of PhD students and postdocs of the Mathematics Department to support the conference organising committee. We organised a mentoring programme, a social event for students and postdocs, a daily newsletter, and a WhatsApp community, and compiled recommendations of local activities.	
Co-organiser of the Antipodeal Dynamics Workshop	2023
I co-organised the virtual workshop with Prof. Claire Postlethwaite, and Prof. Peter Ashwin and Muhammed Fadera (both University of Exeter, UK), which featured invited presentations by Prof. Krasimira Tsaneva-Atanasova, Assoc. Prof. David Simpson, and Assist. Prof. Alexandre Rodrigues, and a virtual poster session.	
Reviewer for Nonlinearity and Journal of Nonlinear Science	2023 – present

Professional development

Academic Career Advancement Programme (ACAP)	2024
A year-long course run by the School of Graduate Studies at the University of Auckland for 20 doctoral candidates, featuring 13 three-hour sessions covering different aspects of an academic career. Ten of these sessions were organised by ACAP participants in pairs. I co-organised a seminar on the <i>Broader Dissemination of Research</i> , which included a one-hour workshop with Prof. Neal Curtis on using ideas from comics to design posters, and a talk by Prof. Cliona Ní Nhurchú on engaging with communities, industry, and government.	
Department of Mathematics Tutor Training	2022
A three-hour workshop run by the Department, focussing on the fundamentals of interacting with students in tutorials, being an engaging and effective tutor, and cultural sensitivity.	

Research visits

School of Mathematics, University of Leeds, UK	Jun. 5 – Jul. 7, 2023
Visited Prof. Alastair Rucklidge to work on understanding the continuity of projected maps of heteroclinic networks in $\mathbb{R}^4$.	
Centre of Mathematics, University of Porto, Portugal	Jun. 19 – 23, 2023
Visited Prof. Sofia Castro to work on various problems concerning the dynamics of trajectories near heteroclinic networks in $\mathbb{R}^4$.	

Presentations

Analysing dynamics near heteroclinic networks with a projected map. <i>Minisymposium on Dynamics Near Homoclinic and Heteroclinic Structures, SIAM Conference on Applications of Dynamical Systems</i> . Denver, CO, USA	May. 2025
Analysing dynamics near heteroclinic networks with a projected map. <i>Australia and New Zealand Industrial and Applied Mathematics Conference</i> . Coffs Harbour, NSW, Australia.	Feb. 2025
Analysing dynamics near heteroclinic networks with a projected map. <i>Special session on Ergodic Theory and Dynamical Systems, Joint Meeting of the NZMS, AustMS, and AMS</i> . University of Auckland, NZ.	Dec. 2024

Travelling waves in spatially-extended models of cyclic competition. <i>Pattern Formation Workshop</i> . University of Auckland, NZ.	Dec. 2024
Dynamics near heteroclinic networks. <i>Applied Mathematics Unit Doctoral Talks</i> . University of Auckland, NZ.	Oct. 2024
Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system. <i>Australia and New Zealand Industrial and Applied Mathematics Conference</i> . Hahndorf, SA, Australia.	Feb. 2024
Switching and cycling near heteroclinic networks as a piecewise smooth dynamical system. <i>New Zealand Mathematical Society Colloquium</i> . Victoria University of Wellington, NZ.	Dec. 2023
Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition. <i>New Zealand Mathematics and Statistics Postgraduate Conference</i> . Wellington, NZ.	Nov. 2023
Dynamics near heteroclinic cycles and networks. <i>Leeds Applied Nonlinear Dynamics Seminar</i> . University of Leeds, UK.	Jun. 2023
Dynamics near heteroclinic cycles and networks. <i>Centro de Matemática da Universidade do Porto Seminar</i> . University of Porto, Portugal.	Jun. 2023
Switching near heteroclinic networks as a piecewise-smooth dynamical system. <i>Minisymposium on Long-Term Dynamics: Attraction and Stability Properties of Invariant Sets, SIAM Conference on Applications of Dynamical Systems</i> . Portland, OR, USA.	May. 2023
Switching near heteroclinic networks as a piecewise-smooth system. <i>Sydney Dynamics Group Workshop: Dynamics in NZ</i> . Karioitahi, NZ.	Nov. 2022
Switching near heteroclinic networks as a piecewise-smooth system. <i>Applied Mathematics Unit Doctoral Talks</i> . University of Auckland, NZ.	Oct. 2022
Stability and cycling near heteroclinic cycles and networks. <i>Department of Mathematics Student Research Conference</i> . University of Auckland, NZ.	Jun. 2022
Stability and cycling near heteroclinic cycles and networks. <i>Department of Mathematics Postgraduate Seminar</i> . University of Auckland, NZ.	Jun. 2022
Spatially-extended models of competition between four species. <i>Department of Mathematics Postgraduate Mid-year Talks</i> . University of Auckland, NZ.	Jul. 2021
A spatially-extended model of competition between four species. <i>Department of Mathematics Student Research Conference</i> . University of Auckland, NZ.	Jun. 2021
Heteroclinic networks in partial differential equations. <i>Department of Mathematics Summer Research Conference</i> . University of Auckland, NZ.	Feb. 2021

Posters

Travelling waves and heteroclinic networks in spatially-extended models of cyclic competition. <i>New Zealand Mathematical Society Colloquium</i> . Victoria University of Wellington, NZ.	Dec. 2023
Switching near heteroclinic networks as a piecewise-smooth dynamical system. <i>Antipodeal Dynamics Workshop</i> . University of Exeter, UK and University of Auckland, NZ.	Oct. 2022

Conferences and workshops attended

Gene Golub SIAM Summer School, on <i>frontiers in multi-dimensional pattern formation (Upcoming)</i>	Aug. 2025
SIAM Conference on Applications of Dynamical Systems	May 2025
Australia and New Zealand Industrial and Applied Mathematics Conference	Feb. 2025
Joint Meeting of the NZMS, AMS, and AustMS	Dec. 2024

Pattern Formation Workshop (Satellite meeting to the Joint Meeting of the NZMS, AMS, and AustMS.)	Dec. 2024
Australia and New Zealand Industrial and Applied Mathematics Conference	Feb. 2024
New Zealand Mathematics Research Institute Summer School, on <i>mathematical neuroscience</i> (Received the Kalman Summer Scholarship to support attendance.)	Jan. 2024
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)	Dec. 2023
New Zealand Mathematical Society Colloquium	Dec. 2023
New Zealand Mathematics and Statistics Postgraduate Student Conference	Nov. 2023
SIAM Conference on Applications of Dynamical Systems	May 2023
New Zealand Mathematics Research Institute Summer School, on <i>groups, geometry, and dynamics</i>	Jan. 2023
Sydney Dynamics Group Workshop: Dynamics in NZ	Nov. 2022
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)	Oct. 2022
Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)	Feb. 2022
Antipodeal Dynamics Workshop, University of Exeter and University of Auckland (Virtual)	Oct. 2021
SIAM Conference on Applications of Dynamical Systems (Virtual)	May 2021
Australia and New Zealand Industrial and Applied Mathematics Conference (Virtual)	Feb. 2021

Professional affiliations

New Zealand Mathematical Society
Australia New Zealand Industrial and Applied Mathematics
Society for Industrial and Applied Mathematics
 Student member

Employment history

Department of Mathematics, Faculty of Science, University of Auckland

Graduate Teaching Assistant

Jul. 2022 — Jun. 2024

Research Assistant

Nov. 2021 — Jun. 2022

38 Princes St, Auckland Central, Auckland, 1010, NZ

Department of Engineering Science and Biomedical Engineering, Faculty of Engineering, University of Auckland

Teaching Assistant

Mar. 2020 — Nov. 2020

70 Symonds Street, Grafton, Auckland, 1010, NZ

SUEZ Smart Solutions Ltd.

Nov. 2019 — Mar. 2020

Assistant Optimisation Engineer

AMI House, 63 Albert St, Auckland Central, Auckland, 1010, NZ

Watercare Services Ltd.

Oct. 2018 — Nov. 2019

Undergraduate Engineer

73 Remuera Road, Remuera, Auckland, 1050, NZ